

Identifying drivers of change during the recent two decades in the Visayan Sea using a trophic modeling approach (Supplementary Material)

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Table S 1 Description and sources of inputs for the Visayan Sea baseline Ecopath model parameters

	Group name	Biomass in habitat area	P/B	Q/B	Diets	Catch
1	Seabirds	Estimated by Ecopath	other model (Ainsworth et al., 2006)	other model (Ainsworth et al., 2006)	other model (Ainsworth et al., 2006)	--
2	Sharks (juvenile/young adult)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=0.61$, after Pauly (1980)	original input: 7.13/yr, the weighted mean of empirical estimates of Q/B of representative species after Pauly et al. (1990)	Small sharks (Pitcher et al., 2002)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
3	Rays	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=0.71$, after Pauly (1980)	original input: 7.77/yr, the weighted mean of empirical estimates of Q/B of representative species after Pauly et al. (1990)	Dasyatidae (Kulbicki et al., 2005)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
4	Pelagic predator fish (i)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=1.57$, after Pauly (1980)	original estimate: 21.21/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Auxis</i> spp., <i>Scomberomorus commerson</i> (Bachok et al., 2004)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
5	Pelagic predator fish (a)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=1.12$, after Pauly (1980)	original input: 14.28/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Scomberomorus commerson</i> (Bachok et al., 2004)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
6	Anchovies, sprats	Estimated by Ecopath	initial input based on empirical estimate of $M=2.61$, after Pauly (1980)	original input: 22.40/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Engraulidae (Hajisamae et al., 2004, Milton et al., 1990)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

7	Sardines (i)	Estimated by Ecopath	<i>Sardinella</i> spp. in the Visayan Sea (Armada, 2004, Lanzuela et al., 2018)	original estimate: 21.57/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	juvenile <i>Sardinella</i> spp. in the Visayan Sea (Linares, 2020)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
8	Sardines (a)	based on the relationship $F=Y/B^i$	<i>Sardinella</i> spp. in the Visayan Sea (Armada, 2004, Lanzuela et al., 2018)	original input: 19.07/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Sardinella</i> spp. in the Visayan Sea (Mequila and Campos, 2007, Linares, 2020)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
9	Mackerels (i)	Estimated by Ecopath	<i>Rastrelliger</i> spp. in the Visayan Sea (Guanco et al., 2009, Guancho, 1991, Mesa, 2019)	original estimate: 23.76/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	food items for juvenile <i>Rastrelliger kanagurta</i> reported in Fishbase (Froese and Pauly, 2020)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
10	Mackerels (a)	based on the relationship $F=Y/B^i$	<i>Rastrelliger</i> spp. in the Visayan Sea (Guanco et al., 2009, Guancho, 1991, Mesa, 2019)	original input: 19.54/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Rastrelliger</i> sp. in the Visayan Sea (Mequila and Campos, 2007)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
11	Scads (i)	Estimated by Ecopath	<i>Decapterus</i> spp. in the Visayan Sea (Guanco et al., 2009, Mesa, 2019)	original estimate: 21.16/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	juvenile <i>Selaroides leptolepis</i> (Yamashita et al., 1987), juvenile <i>Decapterus</i> (Russell, 1983)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
12	Scads (a)	based on the relationship $F=Y/B^i$	<i>Decapterus</i> spp. in the Visayan Sea (Guanco et al., 2009, Mesa, 2019)	original input: 20.27/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	<i>Selaroides leptolepis</i> in the Visayan Sea (Mequila and Campos, 2007); <i>Decapterus</i> spp. (Tiews et al., 1971)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

13	Soft-bottom demersal fish (> 30 cm)	Visayan Sea trawl surveys (Aprieto and Viloso, 1979, Armada, 2004, Ampoyos-Arinque, 2018)	Synodontidae in the Visayan Sea (Mesa, 2019)	original input: 10.36/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Synodontidae in the Visayan Sea (Mequila and Campos, 2007)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
14	Soft-bottom demersal fish (< 30 cm)	Visayan Sea trawl surveys (Aprieto and Viloso, 1979, Armada, 2004, Ampoyos-Arinque, 2018)	Mullidae and Nemipteridae in the Visayan Sea (Mesa, 2019)	original input: 16.62/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Leicognathidae, Mullidae and Nemipteridae in the Visayan Sea (Mequila and Campos, 2007)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
15	Reef-associated and inshore predator fish (1)	based on the relationship $F=Y/B^i$	initial input based on empirical estimate of $M=1.16$, after Pauly (1980)	original input: 11.32/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Carangidae, Rachycentridae, Sphyraenidae (Bachok et al., 2004, Blaber et al., 1990, Kulbicki et al., 2005, Salini et al., 1994)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
16	Reef-associated and inshore predator fish (2)	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=1.08$, after Pauly (1980)	original input: 11.06/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Haemulidae, Lethrinidae, Lutjanidae, Serranidae (Blaber et al., 1990, Kulbicki et al., 2005, Bachok et al., 2004, Salini et al., 1994)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
17	Reef-associated zoobenthos feeding fish	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=1.98$, after Pauly (1980)	original input: 13.89/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Apogonidae, Labridae, Nemipteridae, Pomacentridae (Sano et al., 1984, Pamintuan and Aliño, 1994, Salini et al., 1994)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

18	Reef-associated grazers/herbivorous fish (1)	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=1.46$, after Pauly (1980)	original input: 34.00/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Scarridae (Ochavillo et al., 1992)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
19	Reef-associated grazers/herbivorous fish (2)	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=2.32$, after Pauly (1980)	original input: 43.71/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Siganidae (Adhiambo, 2009, Sano et al., 1984)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
20	Reef-associated planktivorous fish	fish visual census (Campos et al., 2020c, Campos et al., 2020a, Campos et al., 2020d, Campos et al., 2020b)	initial input based on empirical estimate of $M=1.96$, after Pauly (1980)	original input: 18.04/yr, the weighted mean of empirical estimates of Q/B of representative species after Palomares and Pauly (1998)	Caesionidae, Pomacentridae (Sano et al., 1984)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
21	Jellyfish	Estimated by Ecopath	other models (Opitz, 1996, Ainsworth et al., 2006)	other models (Opitz, 1996, Ainsworth et al., 2006)	other models (Opitz, 1996, Ainsworth et al., 2006)	--
22	Octopus	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	Octopus (Pauly et al., 1993a)	<i>Octopus</i> spp. (Ambrose, 1984, McQuaid, 1994, Opitz, 1996)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
23	Squids	Estimated by Ecopath	<i>Uroteuthis</i> spp. in the Visayan Sea (Tajolosa, 2011)	Squids (Pauly et al., 1993a)	<i>Uroteuthis</i> spp. (Islam et al., 2018, Tehseen et al., 2019)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
24	Marine crabs	Estimated by Ecopath	<i>Portunus pelagicus</i> in the Visayan Sea (Ingles, 1996, Mesa et al., 2017, Romero, 2009)	Portunidae (Pauly et al., 1993a)	<i>Portunus</i> spp. (de Lestang et al., 2000, Kunsook et al., 2014, Williams, 1982, Ikhwanuddin et al., 2014)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
25	Shrimps and prawns	Estimated by Ecopath	other models (Armada and Bacalso, 2004, Armada et al., 2003)	Penaeidae (Pauly et al., 1993a)	Penaeidae (Tiews et al., 1968, Marte, 1980)	fisheries monitoring and assessment 1996-97 (Armada, 2004)

26	Molluscs	Estimated by Ecopath	exploited scallops in the Visayan Sea (del Norte-Campos and Villarta, 2008, Cabiles et al., 2019)	other models (Alino et al., 1993, Opitz, 1996)	Gastropods and bivalves (Lin et al., 2006, Opitz, 1996, Gabral-Llana, 1986)	fisheries monitoring and assessment 1996-97 (Armada, 2004)
27	Other epibenthos	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	Echinoderms, (Pauly et al., 1993a)	various references (Alino et al., 1993, Opitz, 1996)	--
28	Worms, infauna	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	other models (Alino et al., 1993, Opitz, 1996)	Polychaetes (Fauchald and Jumars, 1979)	
29	Sessile benthos	Estimated by Ecopath	other models (Alino et al., 1993, Opitz, 1996)	other models (Alino et al., 1993, Opitz, 1996)	various references (Pitcher et al., 2002, Alino et al., 1993)	--
30	Zooplankton	Estimated by Ecopath	other models (Bundy and Pauly, 2001, Alino et al., 1993)	other models (Bundy and Pauly, 2001, Alino et al., 1993)	various references (Alino et al., 1993, Guarín, 1991, Campos, 2003)	--
31	Phytoplankton	Chlorophyll-a concentrations extracted for the Visayan Sea area ⁱⁱ	other models (Bundy and Pauly, 2001, Alino et al., 1993)	other models (Bundy and Pauly, 2001, Alino et al., 1993)	--	--
32	Benthic autotrophs	Estimated by Ecopath	other models (Bundy and Pauly, 2001, Alino et al., 1993)	other models (Bundy and Pauly, 2001, Alino et al., 1993)	--	--
33	Detritus	Empirical estimation after Pauly et al. (1993b)	--	--	--	--

ⁱ where Y is the estimated annual catch (t/km²/year) and F is the fishing mortality rate (/year) initially assumed to be $\leq$ the natural mortality rate M

ⁱⁱ https://coastwatch.pfeg.noaa.gov/erddap/griddap/NCEP_Global_Best.html (accessed 2020); chlorophyll-a to phytoplankton biomass conversion factors from Opitz (1996)

Table S 2 Diet composition matrix of the Visayan Sea baseline Ecopath model arranged by prey (rows) and predator (columns)

	Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Seabirds	--	1E-04	--	--	--	--	--	--	--	--	--	--	--	--	--	--
2	Sharks, juv-ya	--	0.005	--	--	--	--	--	--	--	--	--	--	--	--	--	--
3	Rays	--	0.05	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4	Pel-pred fish i	--	5E-04	--	--	0.005	--	--	--	--	--	--	--	--	--	0.005	--
5	Pel-pred fish a	--	0.003	--	--	0.003	--	--	--	--	--	--	--	--	--	--	--
6	Anchovies	0.207	--	--	0.195	0.109	0.007	--	--	--	0.001	0.01	0.167	0.08	0.002	0.151	0.011
7	Sardines i	0.267	0.008	--	0.11	0.105	--	--	--	--	--	0.007	0.042	0.066	0.002	0.155	--
8	Sardines a	0.05	0.016	--	0.003	0.065	--	--	--	--	--	--	--	0.002	--	0.022	--
9	Mackerels i	0.196	0.062	--	0.105	0.19	--	--	--	--	--	--	0.029	0.069	--	0.107	--
10	Mackerels a	0.01	0.01	--	--	0.015	--	--	--	--	--	--	--	--	--	0.008	--
11	Scads i	0.095	0.008	--	0.05	0.115	--	--	--	--	--	--	0.008	0.028	--	0.09	--
12	Scads a	0.02	0.01	--	--	0.145	--	--	--	--	--	--	--	--	--	0.003	--
13	Dem fish (>30 cm)	0.01	0.26	--	--	0.025	--	--	--	--	--	--	--	0.008	--	0.01	0.004
14	Dem fish (<30 cm)	0.1	0.29	0.001	0.001	0.035	--	--	--	--	--	--	--	0.238	0.035	0.087	0.004
15	Jacks, barracudas	--	0.007	--	--	0.008	--	--	--	--	--	--	--	--	--	0.005	--
16	Rf-carniv.fish (>30 cm)	--	0.044	--	--	0.003	--	--	--	--	--	--	--	0.002	--	0.008	0.015
17	Rf-zoob-feed.fish (<30 cm)	--	0.059	0.001	--	0.003	--	--	--	--	--	--	--	--	--	0.022	0.358
18	Rf-herbiv.fish	--	0.049	--	--	0.003	--	--	--	--	--	--	--	--	--	0.022	0.029
19	Rabbitfish	--	0.016	--	--	0.003	--	--	--	--	--	--	--	--	--	0.008	0.011
20	Rf-planktiv.fish	--	0.035	--	--	0.005	--	--	--	--	--	--	--	--	--	0.03	0.059
21	Jellyfish	--	--	--	0.001	0.001	--	--	--	--	0.001	--	0.001	0.001	5E-04	0.001	0.011
22	Octopus	--	0.004	--	--	--	--	--	--	--	--	--	--	--	--	0.003	0.023
23	Squids	0.005	0.005	0.002	0.02	0.13	--	--	--	--	--	--	--	0.033	5E-04	0.079	0.013
24	Mar.crabs	0.008	0.055	0.165	--	0.005	--	--	--	--	--	--	--	0.15	0.075	0.018	0.197
25	Shrimps	0.015	0.001	0.24	0.15	0.018	0.001	--	--	--	0.009	0.049	0.25	0.22	0.185	0.114	0.102
26	Molluscs	0.007	0.001	0.301	--	--	--	--	--	--	--	0.016	0.087	0.005	0.095	0.007	0.075
27	Oth.epibenthos	0.005	0.001	0.041	--	--	--	--	--	--	--	--	--	0.004	0.015	0.005	0.056
28	Worms, infauna	0.001	--	0.228	--	--	0.049	0.05	0.153	0.05	0.09	0.135	0.045	0.085	0.383	0.025	0.034
29	Sessile benthos	--	--	0.015	--	--	--	--	--	--	--	--	--	--	0.035	--	--
30	Zoopl	0.005	--	--	0.295	0.01	0.82	0.1	0.488	0.547	0.58	0.619	0.322	0.005	0.02	0.015	--
31	Phytopl	--	--	--	0.07	--	0.103	0.85	0.346	0.344	0.302	0.106	--	--	--	--	--
32	Benth.prim.prod	--	--	--	--	--	--	--	--	--	--	--	--	--	0.075	--	--
33	Detritus	--	--	0.006	--	--	0.02	--	0.013	0.059	0.017	0.058	0.049	0.005	0.077	--	--

(...continued Table S2)

	Prey \ predator	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	Seabirds	--	--	--	--	--	--	--	--	--	--	--	--	--	--
2	Sharks, juv-ya	--	--	--	--	--	--	--	--	--	--	--	--	--	--
3	Rays	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4	Pel-pred fish i	--	--	--	--	--	--	--	--	--	--	--	--	--	--
5	Pel-pred fish a	--	--	--	--	--	--	--	--	--	--	--	--	--	--
6	Anchovies	0.002	--	--	--	--	--	0.12	--	--	--	--	--	--	--
7	Sardines i	--	--	--	--	--	--	0.04	--	--	--	--	--	--	--
8	Sardines a	--	--	--	--	--	--	0.001	--	--	--	--	--	--	--
9	Mackerels i	--	--	--	--	--	--	0.005	--	--	--	--	--	--	--
10	Mackerels a	--	--	--	--	--	--	--	--	--	--	--	--	--	--
11	Scads i	--	--	--	--	--	--	--	--	--	--	--	--	--	--
12	Scads a	--	--	--	--	--	--	--	--	--	--	--	--	--	--
13	Dem fish (>30 cm)	--	--	--	--	--	--	--	--	--	--	--	--	--	--
14	Dem fish (<30 cm)	--	--	--	--	--	--	0.02	0.001	--	--	--	--	--	--
15	Jacks, barracudas	--	--	--	--	--	--	--	--	--	--	--	--	--	--
16	Rf-carniv.fish (>30 cm)	--	--	--	--	--	--	--	--	--	--	--	--	--	--
17	Rf-zoob-feed.fish (<30 cm)	0.011	--	--	--	--	0.01	0.001	0.001	--	--	--	--	--	--
18	Rf-herbiv.fish	--	--	--	--	--	0.008	0.001	--	--	--	--	--	--	--
19	Rabbitfish	--	--	--	--	--	0.003	--	--	--	--	--	--	--	--
20	Rf-planktiv.fish	--	--	--	--	--	0.003	0.001	0.001	--	--	--	--	--	--
21	Jellyfish	--	--	--	--	--	--	--	--	--	--	--	--	--	--
22	Octopus	--	--	--	--	--	0.08	--	--	--	--	--	--	--	--
23	Squids	--	--	--	--	--	0.014	0.005	--	--	--	--	--	--	--
24	Mar.crabs	0.051	--	--	--	--	0.151	0.055	0.036	--	--	--	--	--	--
25	Shrimps	0.107	--	--	--	0.008	0.005	0.088	0.038	0.03	--	0.005	--	--	--
26	Molluscs	0.152	--	0.001	--	--	0.463	0.095	0.25	--	0.002	0.01	--	--	--
27	Oth.epibenthos	0.148	--	--	--	--	0.052	0.055	0.076	--	0.018	0.025	--	--	--
28	Worms, infauna	0.371	--	0.015	--	--	0.18	0.029	0.169	0.03	0.148	0.095	0.087	--	--
29	Sessile benthos	0.069	0.01	--	--	--	0.024	--	0.039	--	0.002	0.008	0.013	--	--
30	Zoopl	0.015	--	0.004	0.98	0.992	--	0.31	--	0.15	0.11	0.014	0.015	0.084	0.15
31	Phytopl	--	--	--	--	--	--	--	--	0.495	0.42	0.05	--	0.2	0.8
32	Benth.prim.prod	0.041	0.98	0.972	--	--	--	0.06	0.098	0.177	0.02	0.38	0.235	0.25	--
33	Detritus	0.033	0.01	0.008	0.02	--	0.007	0.114	0.291	0.118	0.28	0.413	0.65	0.466	0.05

Table S 3 Catch estimates (in t/km²) by Ecogroup and Ecofleet in the Visayan Sea baseline Ecopath model (* have values that are < 0.001).

	Group name	COM-DS	COM-PS	COM-RN	COM-BagN	COM-OT	COM-MWT	COM-MSN	B-DS	B-OT	BeachS	SLift,BagN	BSGNs
1	SBirds	--	--	--	--	--	--	--	--	--	--	--	--
2	Sharks	--	--	--	--	0.002	0.001	--	--	0.000*	--	--	0.000*
3	Rays	0.005	--	--	--	0.016	--	--	--	0.003	--	--	0.001
4	Pel-pred (i)	0.010	0.004	0.000*	--	0.011	--	--	0.001	0.000*	--	--	0.009
5	Pel-pred (a)	0.003	0.003	0.002	--	--	--	--	--	--	--	--	0.000*
6	Anchovies	--	--	--	0.042	--	0.033	0.087	--	--	0.344	0.006	--
7	Sard (i)	0.090	0.140	0.010	0.050	0.062	0.161	0.013	0.016	0.021	0.023	0.005	0.002
8	Sard (a)	0.260	0.905	0.027	0.035	0.004	0.192	0.002	0.012	0.004	--	0.003	0.012
9	Mack (i)	0.308	0.055	0.196	0.010	0.030	0.012	0.003	0.035	0.023	--	--	0.145
10	Mack (a)	0.288	0.250	0.456	--	0.005	--	--	--	--	--	--	0.150
11	Scad (i)	0.181	0.007	0.002	0.013	0.033	0.110	0.158	0.015	0.039	0.001	0.002	0.042
12	Scad (a)	0.451	0.353	0.007	0.011	0.050	--	--	--	0.036	--	--	0.123
13	Dem-fish (>30)	0.052	--	--	0.001	0.096	0.050	--	0.002	0.033	--	--	0.011
14	Dem-fish (<30)	0.850	--	--	0.005	1.090	0.064	0.063	0.130	0.390	0.009	0.023	0.066
15	Rf-pred 1	0.092	--	0.001	0.001	0.005	0.044	0.001	0.044	0.009	0.001	--	0.006
16	Rf-pred 2	0.028	--	--	--	0.027	--	--	0.008	0.013	0.009	--	0.004
17	Rf-zfeeders	0.380	--	--	0.001	0.002	--	--	0.065	0.047	0.014	--	0.073
18	Rf-herbiv 1	0.012	--	--	--	--	--	--	0.000*	--	--	--	0.016
19	Rf-herbiv 2	0.001	--	--	--	--	--	--	0.000*	0.002	0.005	0.003	0.005
20	Rf-planktiv	0.035	--	--	--	--	--	--	0.005	--	--	0.006	0.029
21	Jellies	--	--	--	--	0.000*	--	--	--	--	--	--	--
22	Octopus	--	--	--	--	0.001	--	--	--	0.007	--	--	--
23	Squids	0.037	--	--	0.009	0.099	0.055	0.010	0.027	0.041	0.006	0.007	0.001
24	Crabs	0.005	--	--	--	0.010	--	--	0.000	0.000*	0.001	--	0.005
25	Shrimps	--	--	--	--	0.038	--	--	--	0.059	0.004	--	--
26	Molluscs	--	--	--	--	0.126	--	--	--	0.007	--	--	--
27	Oth.epi	--	--	--	--	0.028	--	--	--	0.003	--	--	--
28	Worms	--	--	--	--	--	--	--	--	--	--	--	--
29	Sess.benth	--	--	--	--	0.041	--	--	--	--	--	--	--
30	Zoopl	--	--	--	--	--	--	--	--	--	--	--	--
31	Phytopl	--	--	--	--	--	--	--	--	--	--	--	--
32	Benth.prod	--	--	--	--	--	--	--	--	--	--	--	--
33	Detritus	--	--	--	--	--	--	--	--	--	--	--	--
	Sum	3.088	1.717	0.701	0.178	1.776	0.720	0.336	0.360	0.737	0.417	0.056	0.700

(...continued Table S3)

	Group name	CrabN	CrabLN	CrabP	DGNs	DrGNs	EGNs	FCor	FishP	SquidP	SHLs	MHLs	Trolls
1	SBirds	--	--	--	--	--	--	--	--	--	--	--	--
2	Sharks	0.000*	--	--	0.002	--	--	--	0.001	--	0.001	--	0.000*
3	Rays	0.000*	--	--	--	--	--	0.000*	0.002	--	--	--	--
4	Pel-pred (i)	--	--	--	0.007	--	--	--	--	--	--	--	0.006
5	Pel-pred (a)	--	--	--	0.005	--	--	--	--	--	0.016	--	0.006
6	Anchovies	--	--	--	0.016	0.001	--	0.011	--	--	--	--	--
7	Sard (i)	--	--	--	0.084	--	0.209	0.004	--	--	--	--	--
8	Sard (a)	--	--	--	0.147	0.006	0.434	0.001	--	--	--	--	--
9	Mack (i)	--	--	--	0.019	--	0.004	--	--	--	0.011	0.004	0.003
10	Mack (a)	--	--	--	0.015	--	0.001	--	--	--	0.001	--	--
11	Scad (i)	--	--	--	0.201	--	0.011	0.000*	--	--	0.009	0.002	--
12	Scad (a)	--	--	--	0.199	--	0.007	--	--	--	0.000	0.001	0.002
13	Dem-fish (>30)	0.003	--	0.000*	0.002	--	--	0.000*	0.006	--	0.013	0.001	--
14	Dem-fish (<30)	0.003	--	0.004	0.001	0.001	0.020	0.039	0.025	0.000*	0.361	0.015	--
15	Rf-pred 1	--	--	--	0.020	0.004	0.002	0.000*	0.003	--	0.015	0.001	0.002
16	Rf-pred 2	0.002	--	--	--	0.000	--	0.000*	0.021	0.000*	0.030	0.000	0.002
17	Rf-zfeeders	--	--	--	--	0.004	--	0.005	0.081	--	0.185	0.002	--
18	Rf-herbiv 1	0.000*	--	--	--	0.000*	--	0.000*	0.011	--	--	0.005	--
19	Rf-herbiv 2	--	--	--	--	0.011	--	0.021	0.005	--	--	--	--
20	Rf-planktiv	--	--	--	0.000*	0.002	--	0.000*	0.014	--	--	0.000*	--
21	Jellies	--	--	--	--	--	--	--	--	--	--	--	--
22	Octopus	--	--	--	--	--	--	--	0.000*	--	--	--	--
23	Squids	--	--	--	0.010	--	--	0.001	0.002	0.071	0.009	0.001	0.009
24	Crabs	0.122	0.056	0.064	--	--	--	0.000*	0.001	--	--	--	--
25	Shrimps	--	--	--	--	0.070	--	0.133	0.001	--	--	--	--
26	Molluscs	--	--	--	--	--	--	--	--	--	--	--	--
27	Oth.epi	0.001	--	--	--	--	--	--	--	--	--	--	--
28	Worms	--	--	--	--	--	--	--	--	--	--	--	--
29	Sess.benth	--	--	--	--	--	--	--	--	--	--	--	--
30	Zoopl	--	--	--	--	--	--	--	--	--	--	--	--
31	Phytopl	--	--	--	--	--	--	--	--	--	--	--	--
32	Benth.prod	--	--	--	--	--	--	--	--	--	--	--	--
33	Detritus	--	--	--	--	--	--	--	--	--	--	--	--
	Sum	0.131	0.056	0.068	0.728	0.099	0.687	0.217	0.172	0.071	0.651	0.031	0.030

(...continued Table S3)

	Group name	BSLLs	Compr	Spear	SqJ	PushN	MPushN	RHS	ScoopN	Glean	BlastF	Total
1	SBirds	--	--	--	--	--	--	--	--	--	--	--
2	Sharks	0.005	--	0.000*	--	--	--	--	--	--	--	0.011
3	Rays	0.005	--	0.003	0.000*	--	--	--	--	--	--	0.036
4	Pel-pred (i)	--	--	--	--	--	--	--	--	--	--	0.048
5	Pel-pred (a)	--	--	--	--	--	--	--	--	--	--	0.035
6	Anchovies	--	--	--	--	--	--	0.013	0.005	--	0.018	0.576
7	Sard (i)	--	--	--	--	--	--	0.030	--	--	0.005	0.925
8	Sard (a)	--	--	--	--	--	--	--	--	--	0.005	2.048
9	Mack (i)	--	--	--	--	--	--	--	--	--	0.001	0.858
10	Mack (a)	0.000*	--	--	--	--	--	--	--	--	0.001	1.167
11	Scad (i)	--	--	--	--	--	--	--	0.000*	--	0.001	0.826
12	Scad (a)	0.002	--	--	--	--	--	--	--	--	--	1.241
13	Dem-fish (>30)	0.020	0.001	--	--	--	--	--	--	--	--	0.292
14	Dem-fish (<30)	0.136	0.000*	--	--	0.004	--	0.015	0.007	0.003	0.018	3.342
15	Rf-pred 1	0.011	0.000*	0.005	--	--	--	--	--	--	0.001	0.270
16	Rf-pred 2	0.082	0.026	0.069	--	--	--	--	--	--	0.004	0.324
17	Rf-zfeeders	0.285	0.002	0.035	--	--	--	0.000*	--	--	0.015	1.195
18	Rf-herbiv 1	--	0.004	0.011	--	--	--	--	--	--	0.007	0.067
19	Rf-herbiv 2	--	0.003	0.005	--	--	--	--	--	--	0.014	0.074
20	Rf-planktiv	0.045	0.000*	0.013	--	--	--	--	--	--	0.025	0.176
21	Jellies	--	--	--	--	--	--	--	--	--	--	0.000
22	Octopus	--	0.000*	0.015	0.001	--	--	--	--	--	--	0.024
23	Squids	--	0.000*	0.065	0.095	--	--	0.001	--	--	--	0.555
24	Crabs	0.002	0.002	0.036	--	0.000*	--	--	--	0.032	--	0.337
25	Shrimps	--	--	--	--	0.014	0.027	0.001	0.007	0.009	--	0.363
26	Molluscs	--	0.002	0.023	--	0.001	--	--	--	0.109	--	0.268
27	Oth.epi	--	0.000*	0.010	--	--	--	--	--	0.019	--	0.061
28	Worms	--	--	--	--	--	--	--	--	--	--	--
29	Sess.benth	--	--	--	--	--	--	--	--	--	--	0.041
30	Zoopl	--	--	--	--	--	--	--	--	--	--	--
31	Phytopl	--	--	--	--	--	--	--	--	--	--	--
32	Benth.prod	--	--	--	--	--	--	--	--	--	--	--
33	Detritus	--	--	--	--	--	--	--	--	--	--	--
	Sum	0.594	0.040	0.290	0.096	0.019	0.027	0.060	0.019	0.172	0.115	15.160

Visualized outputs of the PRE- and POST-BAL diagnostics for the Visayan Sea baseline Ecopath model

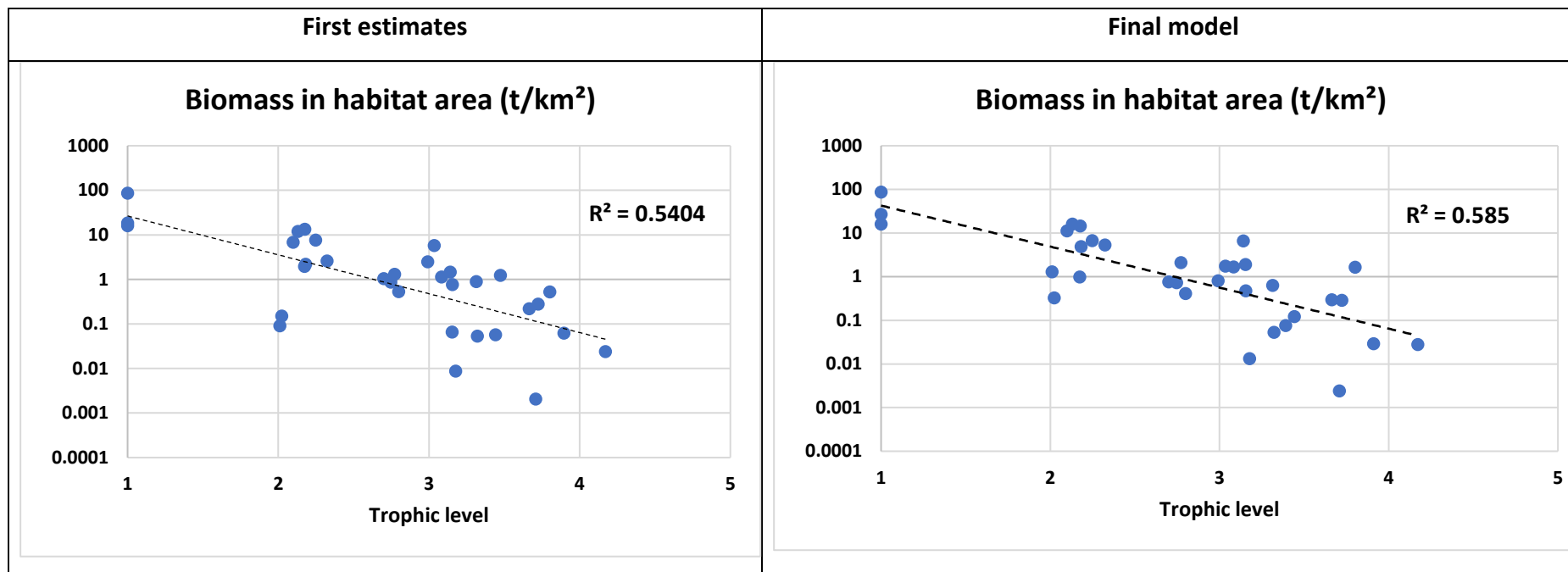


Figure S 1 Biomass (log values) plotted by trophic level of the different groups that make up the Visayan Sea Ecopath model

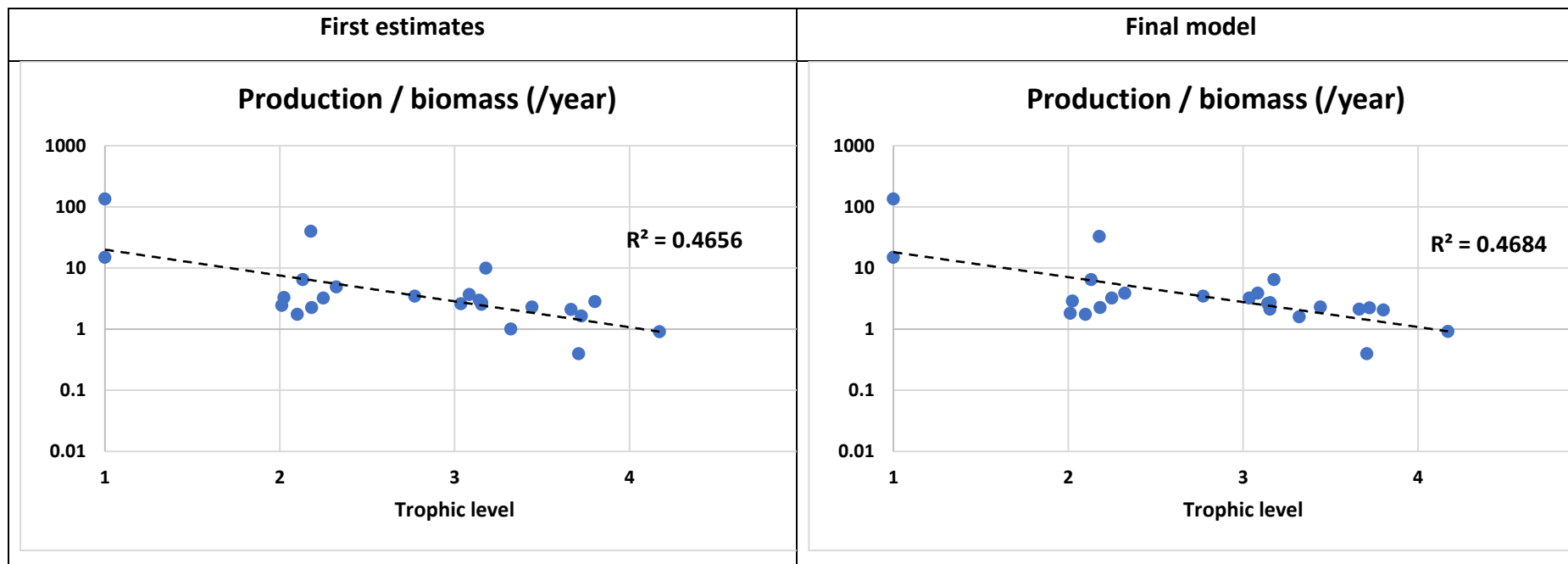


Figure S 2 Production/biomass ratios (log values) plotted by trophic level of the different groups that make up the Visayan Sea Ecopath model

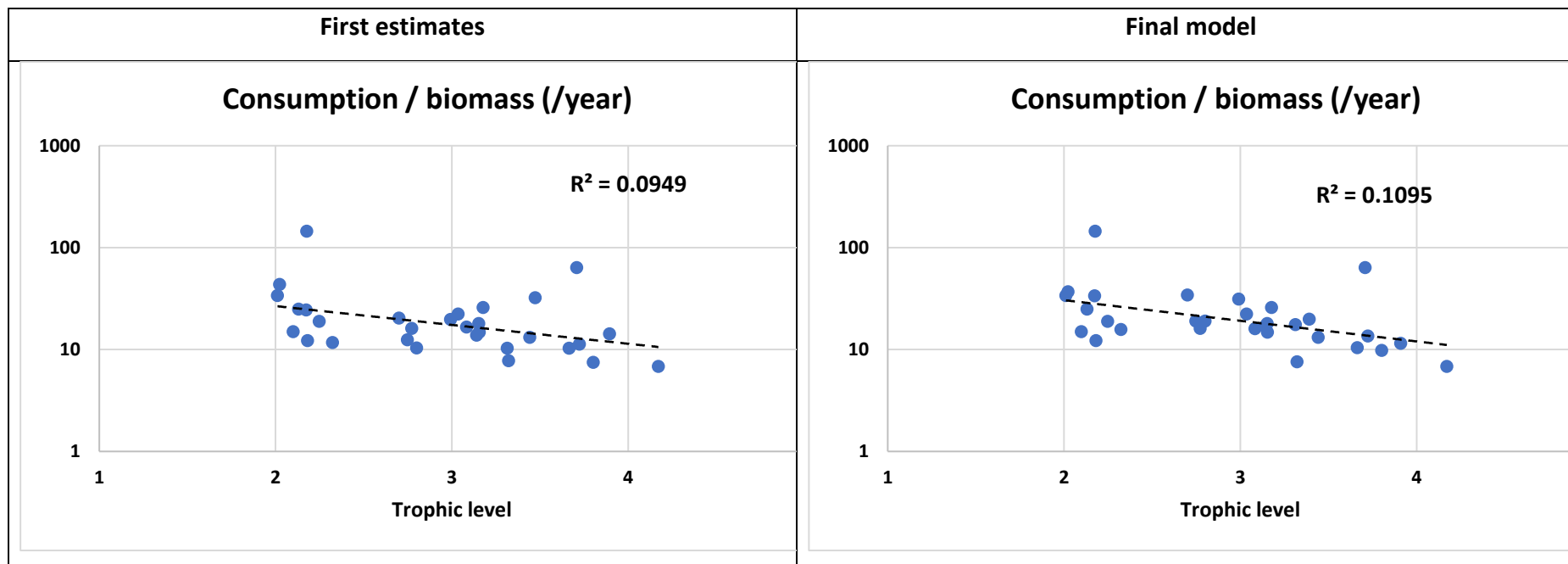


Figure S 3 Consumption/biomass ratios (log values) plotted by trophic level of the different groups that make up the Visayan Sea Ecopath model

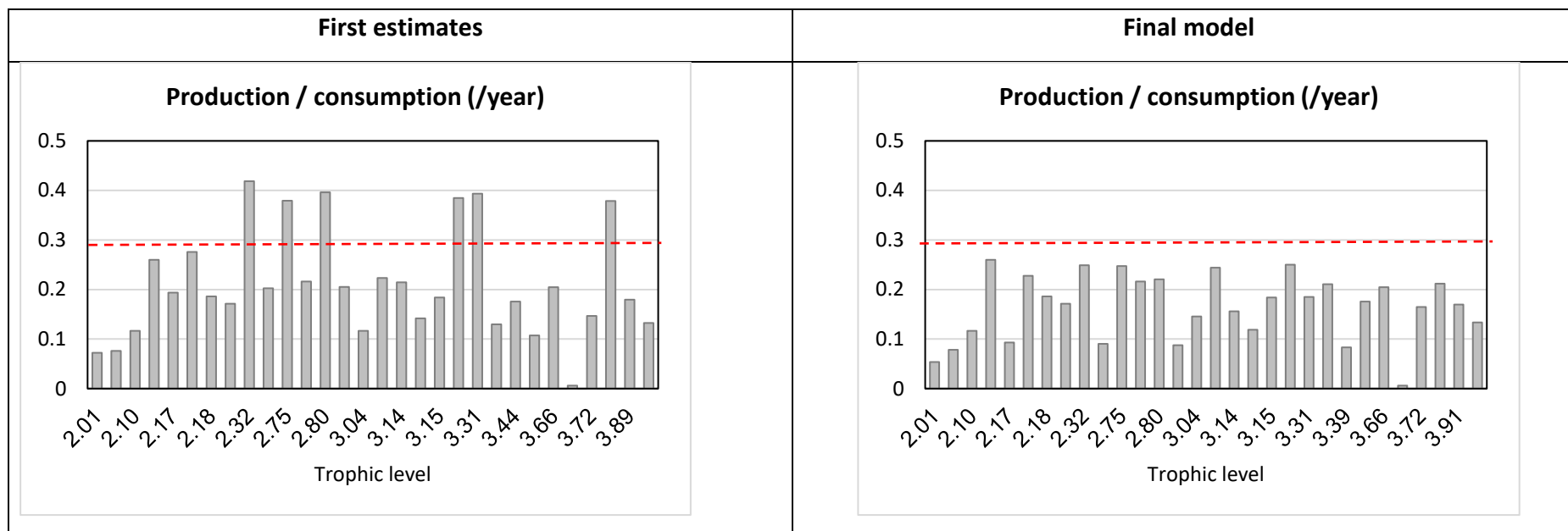


Figure S 4 Production/consumption ratios arranged by trophic level of the different groups that make up the Visayan Sea Ecopath model

Principal Component Analysis (PCA) performed on the 20-year trends of annual species CPUEs from the Visayan Sea, the annually-averaged values of local environmental parameters, and regional climatic indices.

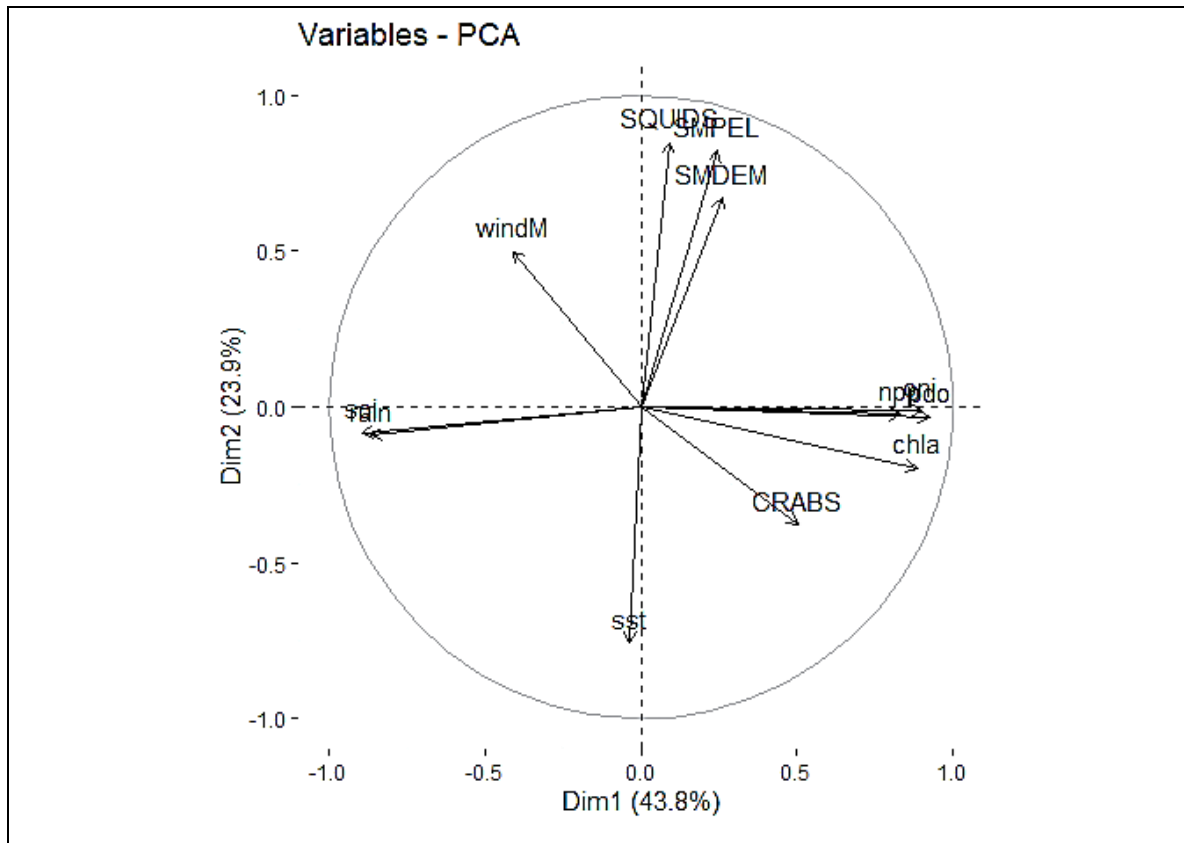


Figure S 5 PCA biplot of the annual CPUEs of small pelagics (SMPEL), small demersal fish (SMDEM), blue swimming crabs (CRABS) and squids (SQUIDS) from the Visayan Sea, the annually-averaged local environmental parameters (sea surface temperature (sst), chlorophyll a (chla), net primary productivity (npp), rainfall (rain), and wind magnitude (windM)), and regional climate indices (Pacific Decadal Oscillation (pdo), Southern Oscillation Index (soi), and Oceanic Niño Index (oni)). Factors 1 and 2 explain 67.7% of the variance.

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